

ArduSwarm: A Self-Contained Micro-Aerial Vehicle Testbed for Experimental Swarm Flight Control

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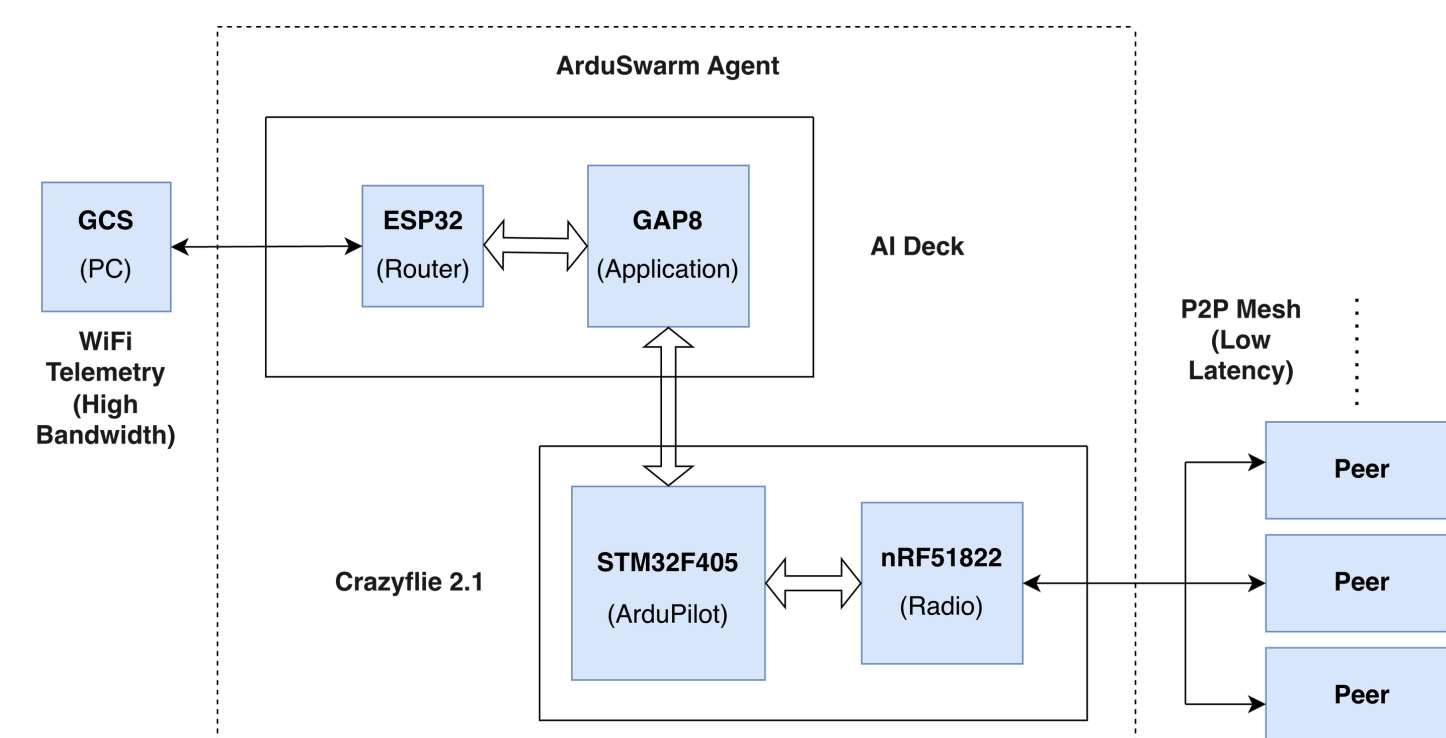
Background

- Micro Air Vehicle (MAV) swarms enable scalable autonomy with redundancy and shift the paradigm from single to many agents per operator [1].
- Existing options force a tradeoff: large outdoor drones (capable but costly) vs micro-drones (accessible but often centralized / tailored flight stacks).
- ArduSwarm** targets lab-scale, self-contained indoor swarming by pairing ArduPilot autonomy within the Bitcraze Crazyflie ecosystem [2][3].

Motivation

- Most micro-swarm work is centralized (hub + shared state), limiting true distributed coordination and introducing a single point of failure [4].
- Fully distributed micro-swarms require: (i) onboard self-state estimation, (ii) low-latency inter-agent comms, (iii) autonomy under tight compute/memory constraints.

Diagram A — Dual Path Communication Pipeline



Contributions:

Infrastructure-free localization: Extended Kalman Filter (EKF) fuses Inertial Measurement Unit (IMU) + optical flow + Time of Flight (ToF) sensor data onboard for 3D position state estimates.

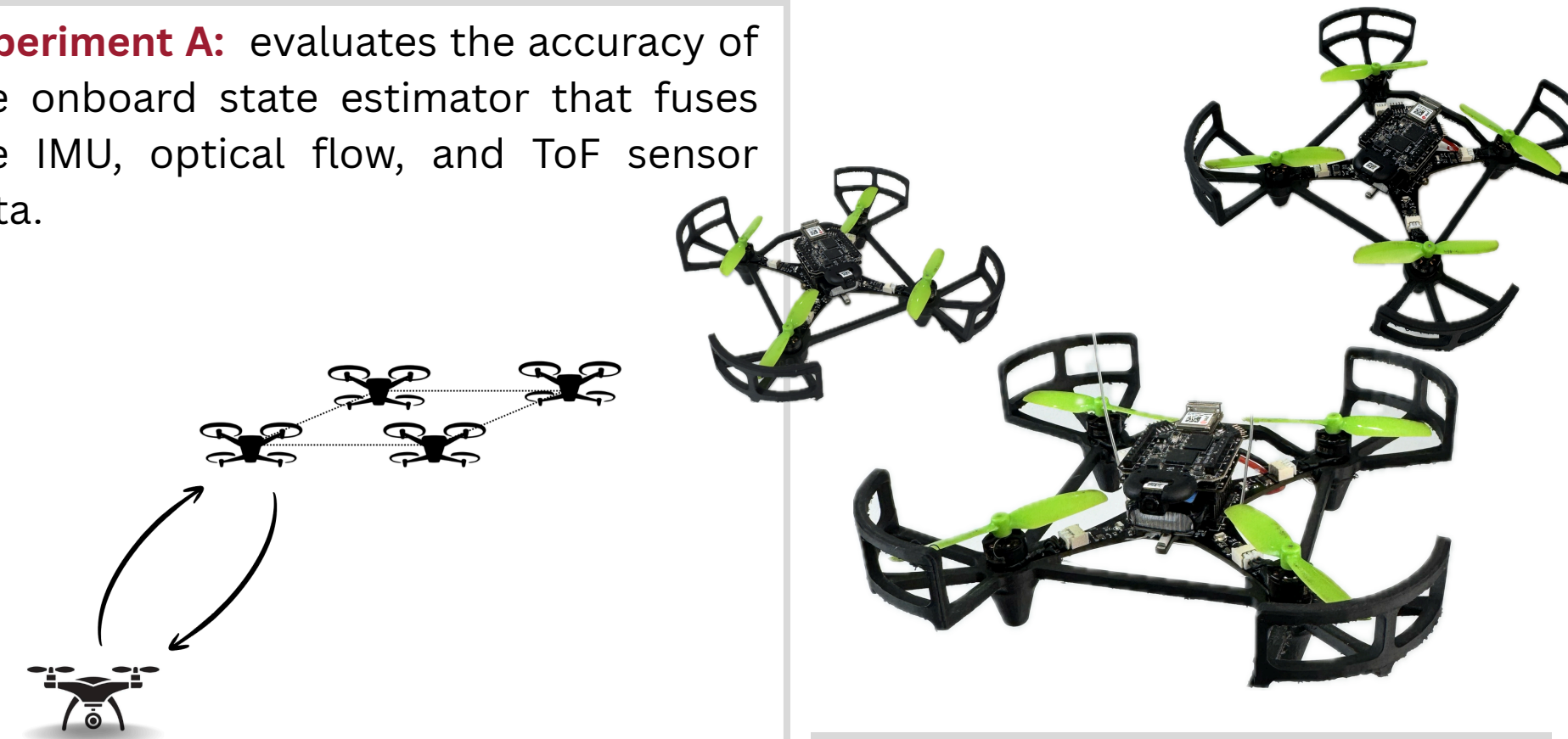
Decentralized network: dual-path comm. pipeline (high-bandwidth telemetry link + low-latency peer to peer (P2P) broadcast links).

Multi-processor Architecture: flight controller for real-time control, companion computer (AI Deck) for coordination/networking.

Methodology and Materials

Two experiments test ArduSwarm's core requirements: (A) infrastructure-free localization and (B) low-latency decentralized communication.

Experiment A: evaluates the accuracy of the onboard state estimator that fuses the IMU, optical flow, and ToF sensor data.

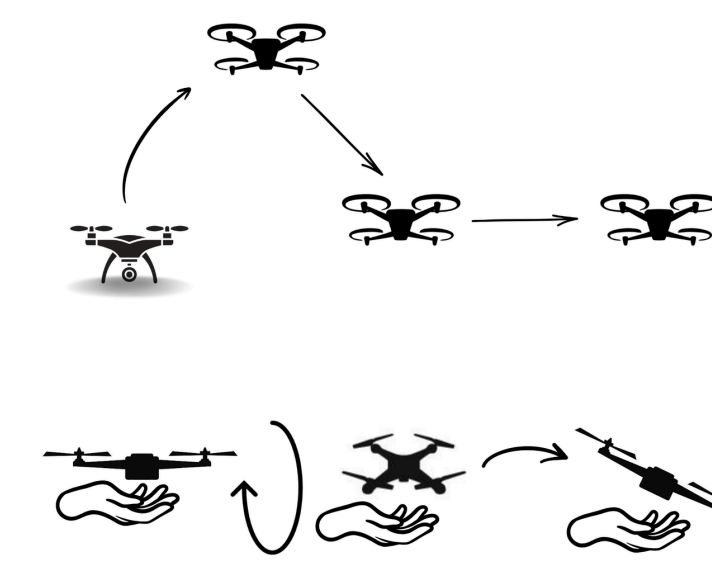


Mission: An agent flies a commanded square trajectory; comparing EKF vs OptiTrack ground truth.
Flight profile: 0.8 m hover → 8 maneuvers × 1 s → land; commanded via RC overrides (1500 μ s ± 200 μ s).

ArduSwarm Components



Experiment B ("Voodoo"): validates the throughput and latency of the P2P network communication pipeline.



Leader (Stationary): An agent held by a human operator streams its attitude (Pitch and Roll) at 10 Hz via the P2P broadcast pipeline.
Follower (Flying): A flying agent listens for P2P broadcasts from the Leader and maps it to a discrete, threshold-based control policy.

Analysis

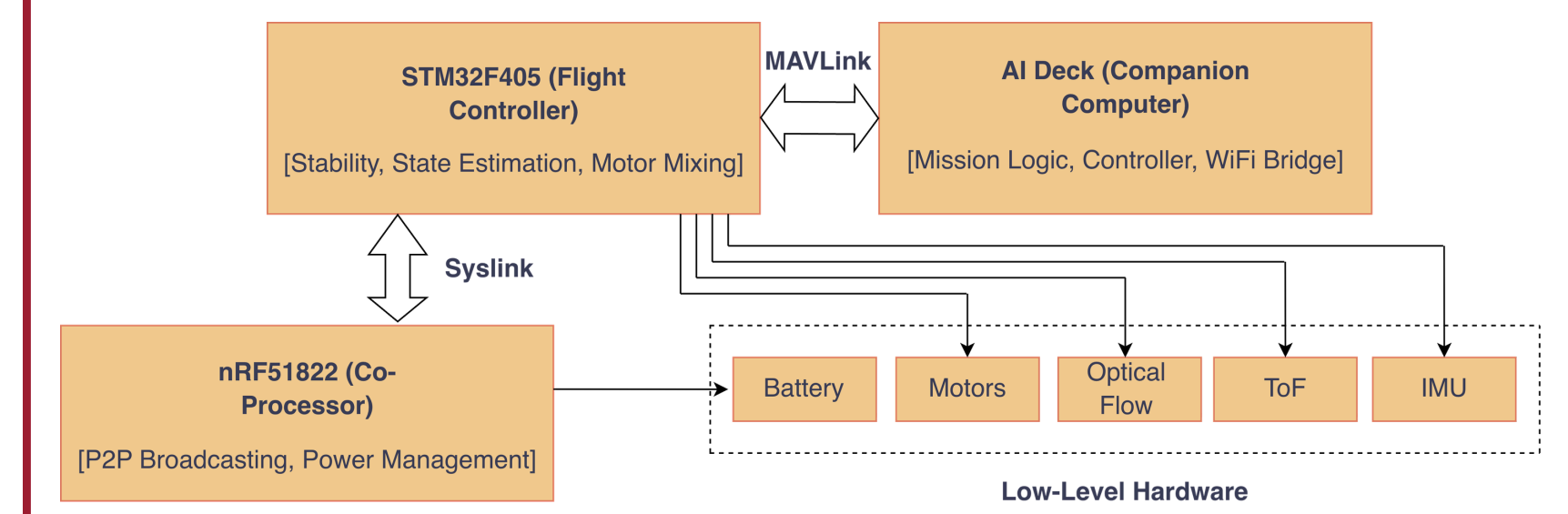
Experiment A — Infrastructure-Free Localization (EKF vs OptiTrack)

- Onboard EKF: IMU + optical flow + ToF → 3D position estimate.
- Log alignment: velocity-correlation time offset ($\Delta t \approx 0.12$ s) + 3 s origin tare.
- Performance: $RMSE_{3D} = 0.353$ m; error dominated by horizontal drift (optical flow), altitude remains tight (ToF).

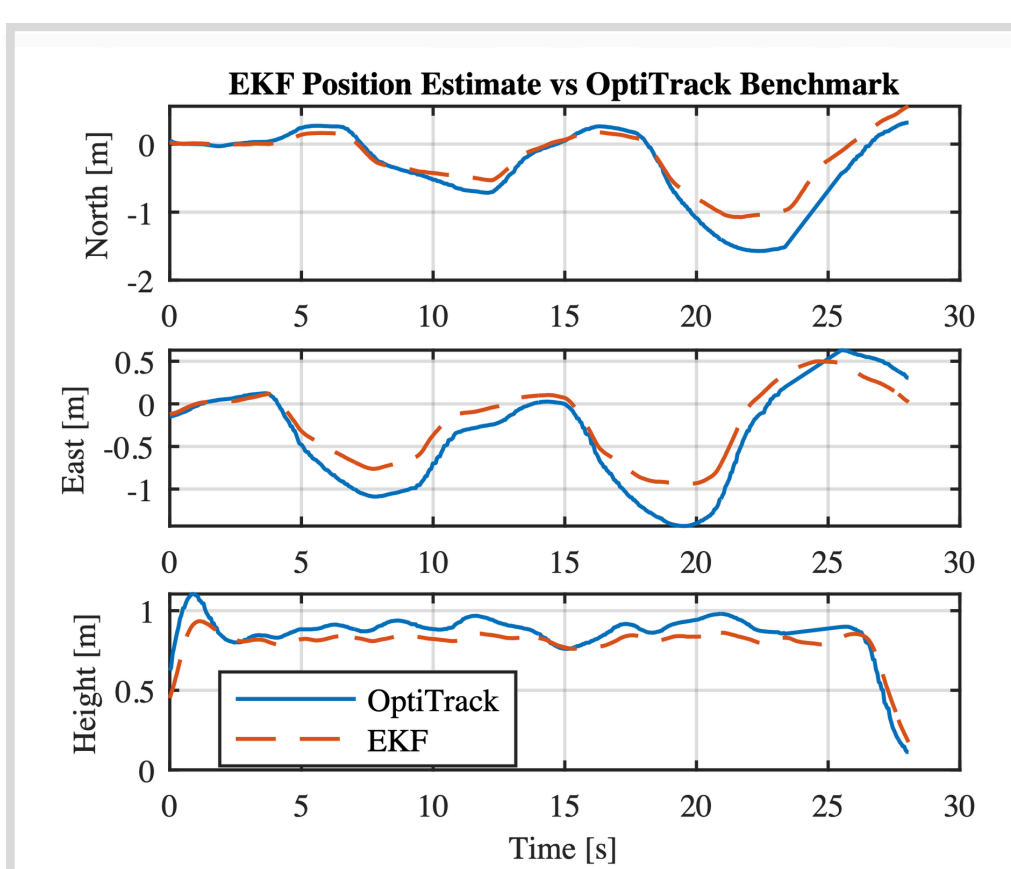
Experiment B — Decentralized P2P Control ("Voodoo")

- Threshold mapping: leader attitude → RC steps (1300/1500/1700 μ s) → follower velocity response.
- Latency definition: $\Delta t = t_{vel} - t_{event}$ (first threshold crossing → sustained response > 0.12 m/s).
- Result: median $\Delta t = 333$ ms end-to-end (E2E), demonstrating direct inter-agent command exchange without a central server.

Diagram B — Multi-Processor Architecture



Results



Experiment A ($RMSE_{3D} = 0.353$ m)

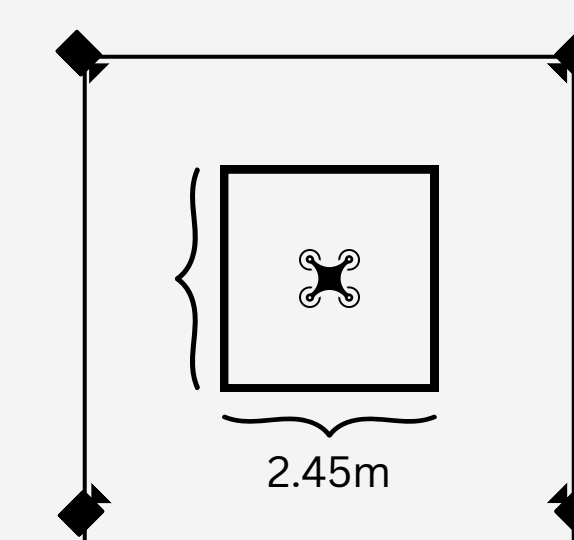
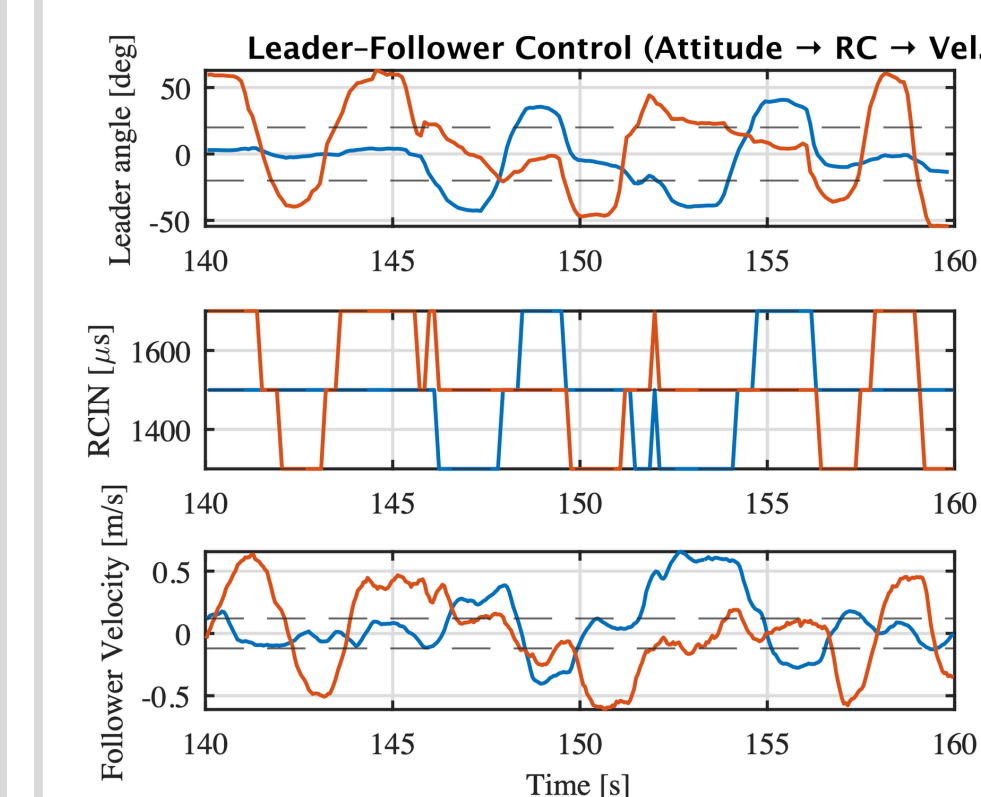


Figure shows the Indoor flight arena, equipped with an OptiTrack motion capture system. Note the OptiTrack system was utilized solely to gather ground-truth data for analysis.



Experiment B (P2P_{E2E} $\Delta t = 0.333$ s)

Conclusion

ArduSwarm brings ArduPilot-class autonomy to micro scale hardware via a multi-processor architecture. We demonstrated infrastructure-free indoor localization (**0.353 m $RMSE_{3D}$**) and decentralized P2P control (**333 ms median E2E latency**). Our platform enables realistic swarm research under micro-drone bandwidth and compute constraints.

Future Work

Future work will use ArduSwarm to study decentralized swarming under realistic bandwidth and compute limits, where agents fuse local sensing and propagate compact state or belief information to support collective decision-making and coordinated action selection.

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References

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Acknowledgements

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